

Anusha Phougat

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SUMMARY

B.Sc. graduate in Microbiology & Chemistry with a strong self-driven transition into Data Science and Machine Learning. Built and deployed end-to-end ML pipelines, interactive web apps, and a production-grade demand forecasting platform featuring LightGBM, Prophet, FastAPI, and Streamlit. Certified in IBM Data Science and Microsoft Azure AI Fundamentals. Brings a unique edge combining life sciences domain knowledge with modern data analytics — well-suited for healthcare analytics, bioinformatics, and research-driven data science roles.

EXPERIENCE

Laboratory Intern — Lamba Family Hospital

Apr 2025 – May 2025

Sonipat, India

- Processed and handled patient samples following standard clinical laboratory protocols, ensuring specimen integrity and biosafety compliance.
- Performed comprehensive urine analysis encompassing physical, chemical, and microscopic examination, contributing to diagnostic workflows.
- Executed microbiological staining procedures including Gram staining for accurate bacterial identification and sample characterization.
- Conducted aseptic culturing of diverse clinical specimens and supported preparation and maintenance of culture media in a sterile environment.
- Maintained rigorous documentation, specimen labeling, and quality control standards aligned with clinical laboratory best practices.

PROJECTS

Retail Demand Forecasting Platform | LightGBM, Prophet, FastAPI, Streamlit, MLflow, Docker | [GitHub](#)

- Built a production-grade retail demand forecasting platform using an ensemble of LightGBM and Facebook Prophet models, improving forecast accuracy over standalone baselines.
- Developed and deployed a FastAPI-based REST API on Render for real-time forecast serving, with Redis caching to reduce repeated prediction latency.
- Designed an interactive Streamlit dashboard deployed on Streamlit Cloud for visual exploration of demand trends, forecasts, and model performance metrics.
- Integrated MLflow for experiment tracking and model versioning, and Evidently AI for automated data drift monitoring in production.
- Containerized the full pipeline with Docker and orchestrated batch retraining workflows using Apache Airflow for scheduled model updates.

Song Recommendation System | Python, Streamlit, scikit-learn, Seaborn, PCA | [GitHub](#)

- Built a hybrid music recommendation engine on the Spotify Tracks Dataset (~114K songs, 114 genres) combining cosine similarity on audio features with a popularity-weighted scoring mechanism.
- Normalized nine audio attributes using MinMaxScaler and performed in-depth EDA with correlation heatmaps, PCA visualizations, and radar charts.
- Deployed as an interactive Streamlit web app supporting search-by-song and search-by-mood modes for real-time personalized recommendations.

Financial Risk Assessment — Loan Approval Predictor | Python, Flask, scikit-learn, HTML/CSS | [GitHub](#)

- Developed an end-to-end loan approval prediction system using classification algorithms trained on applicant financial profiles.

- Built and deployed a Flask web application with an HTML/CSS frontend enabling real-time risk assessment from user-submitted financial inputs.
- Implemented a preprocessing pipeline including missing value handling, feature encoding, and model evaluation to optimize prediction accuracy.

TECHNICAL SKILLS

Languages: Python, SQL

ML & Data: scikit-learn, LightGBM, Prophet, pandas, NumPy, Matplotlib, Seaborn, MLflow, Evidently AI

Web & APIs: FastAPI, Flask, Streamlit

Cloud & DevOps: Microsoft Azure, Docker, Apache Airflow, Git, GitHub, Render

Tools: Microsoft Excel, Jupyter Notebook, VS Code

Concepts: Machine Learning, Time Series Forecasting, EDA, Data Visualization, Recommendation Systems, Classification, REST APIs

Domain Knowledge: Microbiology, Bioinformatics, Clinical Data, Drug Regulatory Affairs

EDUCATION

Mount Carmel College

2023 – 2026

B.Sc. Microbiology & Chemistry (Dual Degree)

Relevant Coursework: Artificial Intelligence, Data Analysis using Excel

CERTIFICATIONS

IBM Data Science Professional Certificate — Coursera / IBM

Covered Python, SQL, data visualization, machine learning, and data science methodology through 10 hands-on courses.

Microsoft Certified: Azure AI Fundamentals (AI-900) — Microsoft

Validated foundational knowledge of AI and machine learning concepts on Microsoft Azure.

Bioinformatics Tools and Techniques — LearnToUpgrade

Learned computational tools for analyzing biological data including sequence analysis and genomic databases.

Drug Regulatory Affairs — LearnToUpgrade

Gained understanding of pharmaceutical regulations, drug approval processes, and compliance frameworks.